NAME: Aniruddh Deepak Nagare
STD-ID: AF05136047
COURSE-ID: ANP-D2217-NCS
COLLEGE: DATTA MEGHE COLLEGE OF ENGINEERING

1. Project Overview

This project demonstrates the complete lifecycle of Virtual LAN (VLAN) configuration and management on a simulated enterprise network. VLANs allow a single physical network to be logically divided into multiple isolated broadcast domains, delivering three core benefits:

- **Security:** Isolate sensitive departments (e.g., Finance, HR) from general users.
- **Performance:** Reduce broadcast traffic by limiting it to relevant segments.
- **Manageability:** Simplify administration by grouping users by function rather than location.

Techniques Covered:

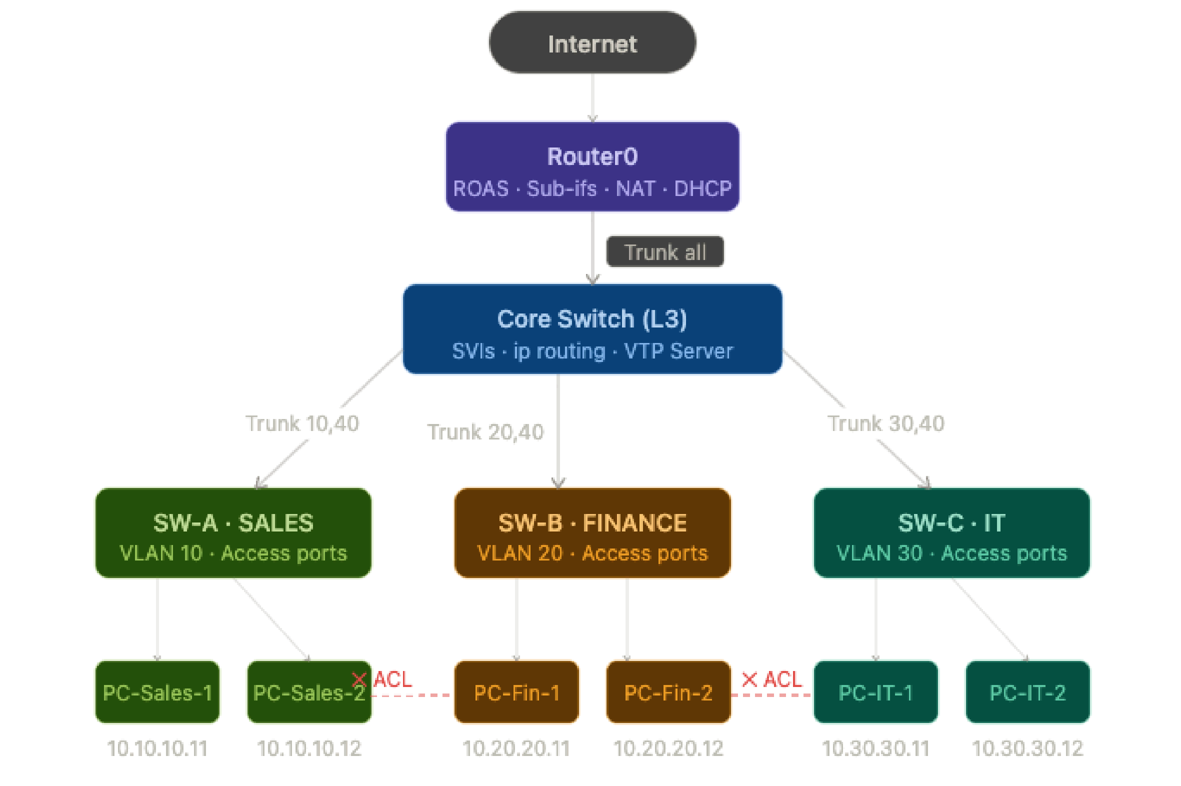
Technique	Description
VLAN creation and naming	Assigning VLAN IDs (1-4094) and human-readable names.
Access port assignment	Binding end-device ports to a single VLAN.
Trunk port configuration	Carrying multiple VLANs over inter-switch links using 802.1Q.
Router-on-a-Stick (ROAS)	Inter-VLAN routing via sub-interfaces on a Layer 3 router.
Layer 3 Switch Routing	Inter-VLAN routing using Switched Virtual Interfaces (SVIs) on a multilayer switch.
VTP (VLAN Trunking Protocol)	Propagating VLAN configurations across multiple switches.
Port Security	Restricting access-port connections by MAC address.

2. Network Topology

- **INTERNET** connection via `Router0`.
- A **Core Switch (Core-SW)** acting as a Layer 3 switch for inter-VLAN routing.
- Three **Access Switches** (`SW-A`, `SW-B`, `SW-C`) connected to the Core Switch via trunk links.
- Separate VLANs for Sales, Finance, and IT departments.

Device Inventory:

Device	Role	Management IP
Router0	Gateway / ROAS	192.168.0.1
Core-SW	Layer 3 Switch	192.168.0.2
SW-A	Access Switch - Sales	192.168.0.10
SW-B	Access Switch Finance	192.168.0.11
SW-C	Access Switch - IT	192.168.0.12



3. VLAN Design

VLAN ID	Name	Subnet	Gateway	Purpose
10	**SALES**	10.10.10.0/24	10.10.10.1	Sales department
20	**FINANCE**	10.20.20.0/24	10.20.20.1	Finance department

30	**IT**	10.30.30.0/24	10.30.30.1	IT/Admin staff
40	**MANAGEMENT**	192.168.0.0/24	192.168.0.1	Switch management
99	**NATIVE**			Untagged trunk traffic

- **Description:** One physical router interface (`Fa0/0`) is divided into logical sub-interfaces, each tagged with a VLAN ID using IEEE 802.1Q encapsulation. The router performs routing between VLANs.
- **Sub-interfaces:**
    - `Fa0/0.10` → VLAN 10 (10.10.10.1/24)
    - `Fa0/0.20` → VLAN 20 (10.20.20.1/24)
    - `Fa0/0.30` → VLAN 30 (10.30.30.1/24)
- **Pros:** Cost-effective (single router interface), easy to implement.
- **Cons:** Single point of failure; physical link can become a bottleneck at high traffic.

Technique 2: Layer 3 Switch with SVIs

- **Description:** A Switched Virtual Interface (SVI) is a virtual Layer 3 interface on the switch tied to a VLAN. The switch itself routes between VLANs without a separate router for inter-VLAN traffic.
- **SVIs (on Core-SW):**
    - VLAN 10 SVI: 10.10.10.1/24
    - VLAN 20 SVI: 10.20.20.1/24
    - VLAN 30 SVI: 10.30.30.1/24
- **Pros:** High throughput (hardware-based routing), no external router needed.
- **Cons:** More expensive than unmanaged switches; requires a multilayer switch.

5. Inter-VLAN Routing

- `C 10.10.10.0/24` [directly connected, Vlan10]
- `C 10.20.20.0/24` [directly connected, Vlan20]
- `C 10.30.30.0/24` [directly connected, Vlan30]
- `S 0.0.0.0/0` [via 192.168.0.1] (Default route to Router0)

Communication Matrix :

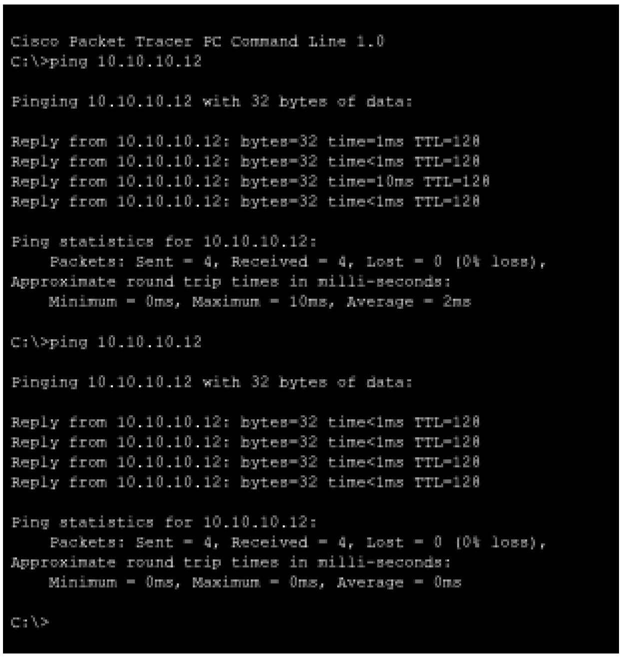
Source VLAN	Destination VLAN	Allowed	Notes
SALES (10)	FINANCE (20)	**No**	Blocked by ACL
SALES (10)	IT (30)	**Yes**	Help desk acc…
FINANCE (20)	IT (30)	**Yes**	IT support

FINANCE (20)	SALES (10)	**No**	Blocked by ACL
IT (30)	All VLANs	**Yes**	Full admin acc…

6. Tests

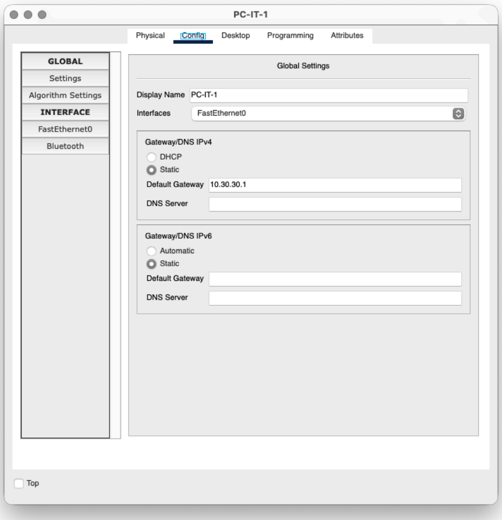
Expected Test Results

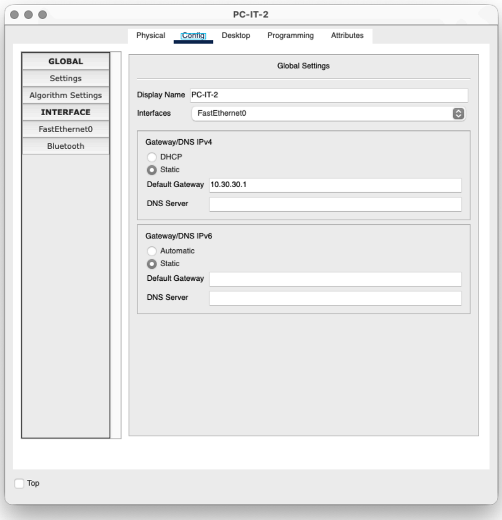
Test	Command	Expected Result
VLAN 10 → VLAN 30	`ping 10.30.30.10 source vlan10`	Success (5/5 packets)
VLAN 10 → VLAN 20	`ping 10.20.20.10 source vlan10`	Fail (ACL deny)
Trunk status	`show interfaces trunk`	Gi0/1 trunking, VLANs 10,20,30,40
SVI status	`show ip int brief`	Vlan10/20/30/40 ~ up/up



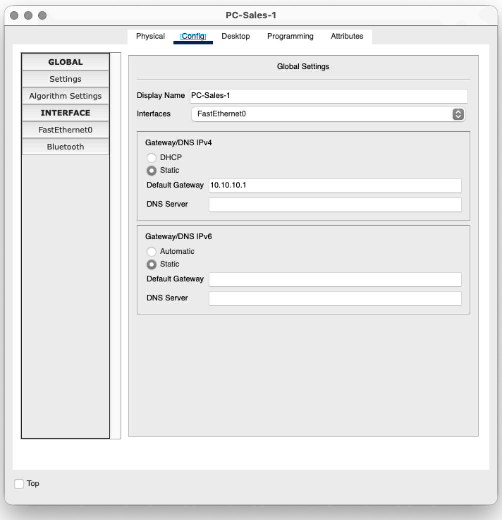
```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 10.10.10.12

Pinging 10.10.10.12 with 32 bytes of data:

Reply from 10.10.10.12: bytes=32 time=1ms TTL=128
Reply from 10.10.10.12: bytes=32 time<1ms TTL=128
Reply from 10.10.10.12: bytes=32 time=10ms TTL=128
Reply from 10.10.10.12: bytes=32 time<1ms TTL=128

Ping statistics for 10.10.10.12:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 10ms, Average = 2ms

C:\>ping 10.10.10.12

Pinging 10.10.10.12 with 32 bytes of data:

Reply from 10.10.10.12: bytes=32 time<1ms TTL=128
Reply from 10.10.10.12: bytes=32 time<1ms TTL=128
Reply from 10.10.10.12: bytes=32 time<1ms TTL=128
Reply from 10.10.10.12: bytes=32 time<1ms TTL=128

Ping statistics for 10.10.10.12:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

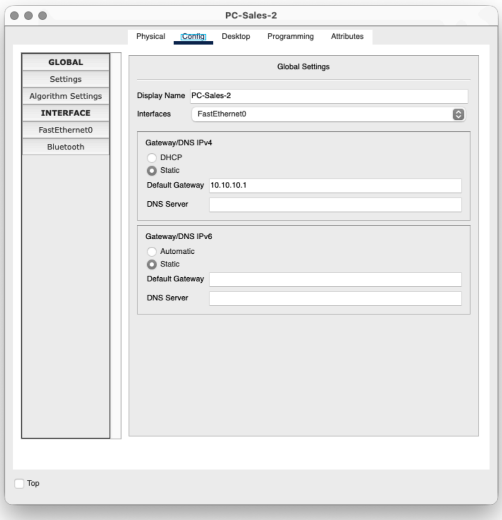
IT


PC-IT-1
Physical
Config
Desktop
Programming
Attributes
GLOBAL
Settings
Algorithm Settings
INTERFACE
FastEthernet0
Bluetooth
Global Settings
Display Name PC-IT-1
Interfaces FastEthernet0
Gateway/DNS IPv4
DHCP
Static
Default Gateway 10.30.30.1
DNS Server
Gateway/DNS IPv6
Automatic
Static
Default Gateway
DNS Server
Top

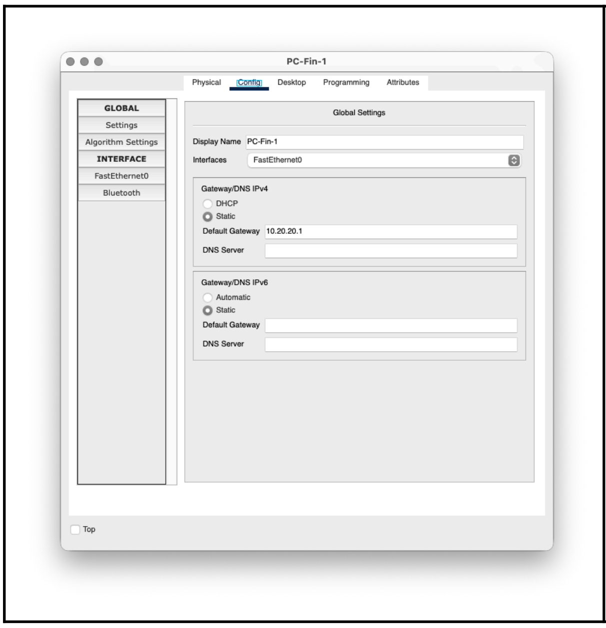

PC-IT-2
Physical
Config
Desktop
Programming
Attributes
GLOBAL
Settings
Algorithm Settings
INTERFACE
FastEthernet0
Bluetooth
Global Settings
Display Name PC-IT-2
Interfaces FastEthernet0
Gateway/DNS IPv4
DHCP
Static
Default Gateway 10.30.30.1
DNS Server
Gateway/DNS IPv6
Automatic
Static
Default Gateway
DNS Server
Top

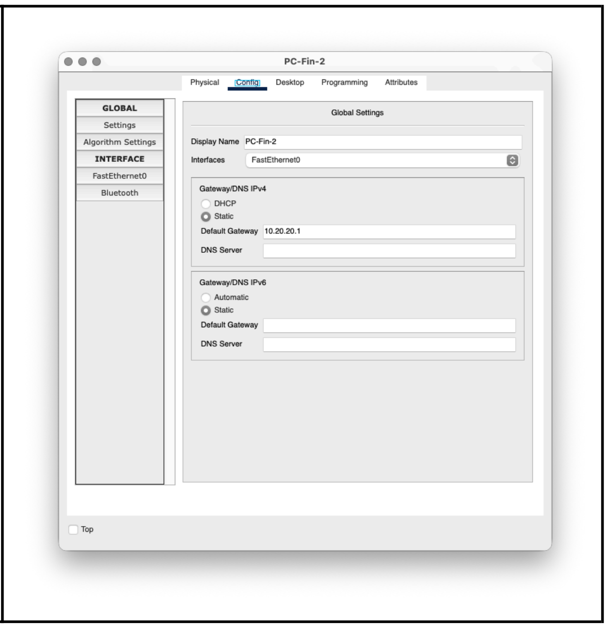
SALES


PC-Sales-1
Physical
Config
Desktop
Programming
Attributes
GLOBAL
Settings
Algorithm Settings
INTERFACE
FastEthernet0
Bluetooth
Global Settings
Display Name PC-Sales-1
Interfaces FastEthernet0
Gateway/DNS IPv4
DHCP
Static
Default Gateway 10.10.10.1
DNS Server
Gateway/DNS IPv6
Automatic
Static
Default Gateway
DNS Server
Top


PC-Sales-2
Physical
Config
Desktop
Programming
Attributes
GLOBAL
Settings
Algorithm Settings
INTERFACE
FastEthernet0
Bluetooth
Global Settings
Display Name PC-Sales-2
Interfaces FastEthernet0
Gateway/DNS IPv4
DHCP
Static
Default Gateway 10.10.10.1
DNS Server
Gateway/DNS IPv6
Automatic
Static
Default Gateway
DNS Server
Top

FIN


PC-Fin-1
Physical
Config
Desktop
Programming
Attributes
GLOBAL
Settings
Algorithm Settings
INTERFACE
FastEthernet0
Bluetooth
Global Settings
Display Name PC-Fin-1
Interfaces FastEthernet0
Gateway/DNS IPv4
DHCP
Static
Default Gateway 10.20.20.1
DNS Server
Gateway/DNS IPv6
Automatic
Static
Default Gateway
DNS Server
Top


PC-Fin-2
Physical
Config
Desktop
Programming
Attributes
GLOBAL
Settings
Algorithm Settings
INTERFACE
FastEthernet0
Bluetooth
Global Settings
Display Name PC-Fin-2
Interfaces FastEthernet0
Gateway/DNS IPv4
DHCP
Static
Default Gateway 10.20.20.1
DNS Server
Gateway/DNS IPv6
Automatic
Static
Default Gateway
DNS Server
Top